ADNAN BAIG Associate Software Engineer (AI/ML)

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PROFILE

Software Engineer specializing in Artificial Intelligence and Machine Learning, with deep expertise in Generative AI and Agentic AI systems. Proven experience in designing and developing intelligent AI agents, conversational interfaces, and scalable backend services. Adept at building production-grade AI solutions, optimizing system performance, and delivering high-impact outcomes in dynamic environments. Strong collaborator with a track record of driving end-to-end execution from concept to deployment.

EDUCATION

Bachelor of Engineering (B.E.), PRMITR, Badnera — 2020 – 2023
82.55% — Amravati Maharashtra

Diploma in Engineering, Government College of Polytechnic — 2017 – 2020
82.26% — Achalpur Maharashtra

PROFESSIONAL EXPERIENCE

Associate Software Engineer (AI/ML), Kairos Technologies — 08/2024 – Present
Hyderabad, India

1. TAP Agents (Kitap Platform)

- Engineered and delivered an advanced AI-powered test automation system within the Kitap platform, leveraging Generative AI and Agentic AI architectures. Designed a multi-agent collaborative framework to autonomously perform web application analysis, feature discovery, and business flow identification.
- Enabled automated generation of manual test cases, BDD scenarios, and browser-based test cases automation, reducing manual testing effort by **70–80%** and eliminating the need for specialized automation expertise. Orchestrated end-to-end test lifecycle automation from application understanding to execution significantly improving testing efficiency, scalability, and release velocity.

2. Data Quality Gateway :

- Contributed to DQG development, focusing on data profiling, cleansing, and large-scale row comparison using Apache Spark. Integrated multi-database support (Snowflake, MySQL, PostgreSQL, Databricks, Excel).
- Built an **AI-powered analysis agent** for cross-database insights via natural language queries. Engineered a **conversational agent** to retrieve metadata and monitor task progress interactively.

Data Science Intern, Innomatics Research Labs — 01/2024 – 04/2024
Hyderabad, India

- Developed and implemented advanced ML models and Generative AI solutions, including NLP techniques such as sentiment analysis and AI-powered applications.
- Gained hands-on experience with data preprocessing, exploratory data analysis, ML model training, evaluation and deployments.

PROJECTS

Test Auto Pilot (Agentic AI Testing System), Kitap — 08/2025 – Present

- Architected and delivered a scalable Agentic AI–driven multi-agent platform enabling end-to-end autonomous testing of web applications using inputs from URLs, BRD documents, Figma designs, Jira user stories, and natural language prompts. Leveraged advanced LLMs and custom orchestration to automate the complete QA lifecycle.
- Designed specialized agents for **overview generation and requirement understanding, feature discovery, and business flow mapping**, along with automated **manual and BDD test case generation and Playwright-based execution**. Extended the platform with advanced capabilities including **visual testing (baseline vs target validation), accessibility testing, smart regression planning, and requirement compliance analysis**.

- Delivered a **~80% reduction in manual testing effort** and **~85% reduction in human intervention**, fully automating the test design, validation, and execution lifecycle—significantly enhancing QA efficiency, scalability, and release velocity.

Data Analytics AI Agents, DQG 11/2024 – Present

- Built an **NLP-powered data analysis agent** that answers complex questions in natural language using both relational database data and requirements documents **enabling deep data insights into data.**
- Implemented **anomaly detection** and automated reasoning over large-scale enterprise data.
- Developed **Apache Spark based row-wise data comparison** system optimized for large datasets using distributed **Spark clusters**.
- Integrated with multiple databases including **MySQL, PostgreSQL, Snowflake, Databricks, and Azure SQL** to ensure cross-platform compatibility.

AI-Powered Face Recognition & PPE/Accident Detection System 09/2024 – 11/2024

- Developed a **face recognition system** capable of training on large datasets to detect and identify individuals by name in real-time.
- Built a **PPE detection module** to monitor compliance with safety gear (helmets, vests, gloves, etc.) in industrial environments.
- Implemented **accident detection features** (e.g., fire detection) using object detection models; integrated **real-time alerting via WhatsApp** to notify administrators of safety breaches.
- Enhanced workplace safety and monitoring using **computer vision, deep learning and AI** techniques.

SKILLS

Technical Skills

- **Programming & Core Technologies:** Python, SQL, Async Programming.
- **AI/ML & Data Science:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Model Fine-tuning, Embeddings, Semantic Search.
- **Backend & API Development:** RESTful APIs (FastAPI), Microservices Architecture, Event-Driven Systems, Asynchronous Processing (Celery), Messaging (RabbitMQ), Redis Caching, Scalable Backend Systems
- **Agentic AI & Frameworks:** LangGraph, LangChain, FastAgent, Google ADK.
- **Data & Vector Databases:** ChromaDB, FAISS, Pinecone (conceptual/working knowledge), PostgreSQL.
- **Libraries & Tools:** Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, Keras, Transformers, OpenCV, Matplotlib, Seaborn, Playwright, Crawl4AI, ChromaDB, Logfire, LLM Instructor.
- **LLM Ecosystem:** OpenAI GPT, Gemini, Anthropic, NVIDIA AI, Hugging Face, Ollama
- **Development & Collaboration Tools:** VS Code, Cursor, Antigravity, Jupyter Notebook, Google Colab
- **Deployment:** Service Deployment, API Serving, Docker (containerization), CI/CD basics, Monitoring & Logging

Soft Skills

Effective verbal and written communication
Presentation skills
Problem solving
Constant self-improvement

CERTIFICATIONS

- Data Science Professional certificate approved by Innomatics Research Labs Hyd.
- Advance ML and Generative AI Internship Program at Innomatics Research Labs Hyd,
- Machine Learnings with Python, powered by IBM and issued by Innomatics Research Labs Hyd.
- Advance Data Analysis Internship Program at Innomatics Research Labs Hyd.

INTERESTS

Tech Enthusiast | Lifelong Learner | Travelling | Foodie

LANGUAGES

English **Hindi** **Urdu** **Marathi**